Predicting Handoff Compatibility for Driving and Gap Crossing of a Single-Track Robot

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Abstract— Crossing a gap does not complete a single-track traversal task if the landing state prevents the ground controller from continuing within the road corridor. We study this handoff through two learned interfaces while keeping the driving and jumping controllers frozen. An action-conditioned model predicts whether a valid handoff occurs and the distribution of the state delivered to the ground controller. A conditional outcome model evaluates whether that controller can complete the remaining task from the predicted state. The planner integrates these predictions over finite route, approach-speed, and preference candidates, with a separate outcome branch for ground-only routes. Supervised fitting uses 1,919 published executions partitioned by starting condition. On 540 development executions, the composed model achieves a completion Brier score of 0.1152 and conditional handoff-position RMSE of 0.132 m. In a subsequent Isaac simulation panel of 30 new starting conditions and three crossed-slot repetitions, however, it completes 15/90 tasks, compared with 33/90 for a direct classifier and 41/90 for a distance rule. All 33 selected detours time out. The results establish a working state-prediction and skill-composition interface, but also show that accurate average prediction does not ensure useful candidate rankings. They identify ground-route scoring and task-conditioned continuation as unresolved requirements for reliable handoff planning.

Index terms: Model learning for control; motion planning; skill composition; single-track robots.

I. Introduction

A single-track robot must maintain dynamic balance while steering toward a goal. A gap introduces a controller transition: ground driving prepares a jump, and the jumping skill returns a moving vehicle to ground control. Reaching the opposite bank is therefore an intermediate event. The state at contact must also allow the next controller to continue within the available road space.

This problem differs from evaluating each skill at its own nominal initial state. Approach commands alter the state entering the jump, and the jump changes the state from which driving resumes. A frozen controller can be competent in isolation yet receive unsuitable states when composed with another skill. Sequential behavior composition provides a foundation for reasoning about such interfaces [1]; here we examine a learned, finite-candidate approximation for a single-track platform.

The research question is whether predicting the distribution of handoff states and their downstream consequences can guide approach decisions. Our method separates the upstream distribution from a conditional full-task outcome predictor. It retains failed approaches as negative handoff examples and does not equate contact, controller switching, or short survival with goal completion. The term world model denotes this restricted predictor of action-conditioned physical outcomes; no visual generation or new low-level reinforcement learning is involved.

The implemented contribution is a two-interface prediction and planning system with explicit state outputs, grouped supervised training, and execution while the vehicle remains in motion. We evaluate prediction separately from online decisions and preserve a direct-success comparator and a distance rule. The present evidence supports executable composition and identifies a candidate-ranking failure. It does not establish superior planning, a certified recoverability boundary, or successful adaptation to new downstream corridors.

II. Related Work

Composing locomotion behaviors. Sequential behavior composition organizes dynamical controllers into larger tasks [1]. For wheeled bipeds, Xin and Vijayakumar combine rolling and contact-changing template models with online planning and whole-body control [2]. Lee et al. integrate learned locomotion and navigation for wheeled-legged robots [3]. Our setting differs in morphology and interface: a single-track vehicle uses an existing ground controller and fixed jumping policy, while the upper layer evaluates their predicted event-level consequences. These precedents motivate composition, rather than establishing a claim to first introducing it.

Learned models for planning. PETS combines probabilistic dynamics with sampled uncertainty propagation [4]. Yang et al. study learned locomotion models and account for planning latency [5]. DayDreamer learns world models for physical robots, including locomotion and navigation [6]. Here, learning is supervised on simulator executions, and the action is a route–skill decision rather than a low-level control sequence. The model predicts the handoff state and conditional continuation outcome; it does not train an imagined policy. A direct success predictor is retained as a comparator: intermediate predictions should be evaluated beyond their ability to assign a final score.

III. Task and System Interface

A. Common-goal traversal

Let $\mathcal{H}$ be a known height map with support mask, $\mathcal{R}$ a finite road set, and $\mathcal{G}$ a shared goal. A task action is

$$a = (r, v^\star, \alpha, \kappa), \quad r \in \mathcal{R}, \quad \kappa \in \{0, 1\}, \tag{1}$$

where $v^\star$ and α specify approach speed and ground-controller preference, and κ selects ground-only traversal or a ground–jump–ground sequence. Every route targets the same region. The robot continues moving during planning; stopping or restoring a saved physical state is not used to resolve a late decision.

The map contains a 0.30-m open gap, 1–4-mm surface stones outside the jump region, and 0.02-m and 0.08-m ridges. The four candidate routes share a destination while providing different approaches or a continuous-ground detour.

Task success requires reaching $\mathcal{G}$ at the declared minimum forward speed, without physical termination or any recorded corridor violation:

$$Y = \mathbf{1}_{\mathcal{G}} \mathbf{1}_{\text{physical valid}} \mathbf{1}_{\text{corridor valid}}. \tag{2}$$

Far-bank contact is intermediate, not success. Bypasses need not generate flight or landing events. Failure endpoints and milestones are retained, so surviving briefly after contact is not conflated with task completion.

B. Frozen controllers and handoff

Ground motion uses the existing STTW H75 controller with ECBC/ESO and zero lower learned residual. Jumping uses a frozen heading-conditioned checkpoint, heading125. H75 adjusts speed/steering references from causal history and path preview above the balance controller and extended-state observer (ESO). Increasing α shifts the trained preference from path following toward speed tracking, with finite secondary costs and unchanged actuator limits. These identities are held fixed in the main study. At the jump-entry line, control switches to the jump policy; this line is distinct from observed takeoff. After far-bank contact, ground control resumes at the next 20-ms boundary. No physical state reset is performed. Controller initialization follows the existing handoff interface rather than assuming interchangeable ground and airborne histories.

The chosen $(v^\star, \alpha)$ controls approach. Postlanding motion uses fixed 1.5 m/s and $\alpha = 0.5$, not separately optimized settings. The planning objective thus concerns the consequences of approach decisions for later continuation. The continuation predictor targets the final goal in (2); it is not a dedicated estimator of stabilization distance.

IV. Handoff-Conditioned World Model

A. Inputs and event semantics

Let V_h denote a handoff that occurs without an earlier road violation. A failed approach or a missing handoff is a negative event label, even when no terminal state can be assigned. For a published candidate a, the model input is

$$\xi = [x_t, o_t, c_t, t, a_{\text{enc}}, H_t] \in \mathbb{R}^{582}, \tag{3}$$

where x_t is the 13-dimensional root state, o_t the 80-dimensional current observation, and c_t the 352-dimensional recorded ground-controller history. The remaining inputs are physical time, an eight-dimensional candidate encoding, and a 128-dimensional known-map crop. Candidate encoding describes the route and its speed, preference, and jump flag; it is not a motor command.

The map crop contains an 8×8 height grid and corresponding support mask sampled from the terrain mesh. The model therefore has access to known geometry, rather than inferred visual terrain. Normalization uses TRAIN statistics only. Unlike the earlier event-chain prototype, this model does not propagate a PCA history vector through successive predicted events.

The conditional output is a 39-dimensional handoff descriptor

$$z_h = [x_h, f_h, t_h], \tag{4}$$

with root state $x_h \in \mathbb{R}^{13}$, latest measured observation frame $f_h \in \mathbb{R}^{25}$, and physical handoff time t_h. Recorded event roots were checked against the corresponding physical transition. The frame is available before switching; it is not a future trajectory observation. This descriptor is an approximation to the closed-loop state, not a serialization of all joints, estimator history, or hidden contact-solver state.

B. Upstream distribution and downstream outcome

Three independently initialized MLPs, each with 128 and 64 SiLU hidden units, predict a diagonal Gaussian and two Bernoulli outputs:

$$F_\theta(\xi) = (\mu_h, \log \sigma_h^2, p_h, p_g). \tag{5}$$

There are 80 scalar outputs: 39 means, 39 log variances, valid-handoff probability p_h, and ground-route completion probability p_g. The ground output shares the trunk; it is a separate output head, not an independently fitted model.

A second three-member ensemble, with 64 and 32 SiLU hidden units, estimates

$$R_\phi(z_h) = P(Y = 1 \mid V_h, z_h, I, \pi_{\text{H75}}), \tag{6}$$

where I is the fixed cold-start initialization rule. H75, its postlanding speed and preference, and the downstream straight-road context are fixed. Thus R_ϕ estimates continuation to the original goal within this context, not a general recovery domain over arbitrary corridor widths, curves, or controller versions. A separate two-second continuation pilot is retained as diagnostic evidence and is not substituted for this full-goal target.

C. Masked supervised training

Let J mark jump candidates and G ground-only candidates. For normalized handoff targets, the upstream

objective is

$$\mathcal{L}_U = \mathrm{BCE}_J(p_h, V_h) + \mathrm{BCE}_G(p_g, Y) + \tfrac{1}{2}\mathbb{E}_{V_h=1}\left[\frac{1}{39}\sum_{d=1}^{39}\left((z_d-\mu_d)^2 e^{-\ell_d}+\ell_d\right)\right], \quad (7)$$

where ℓ_d is log variance clipped to $[-4, 3]$. Missing states are excluded only from the conditional Gaussian loss. They remain classification examples; unsuccessful but valid handoffs train the downstream model with negative goal labels.

The upstream and direct-success networks are fitted for 250 epochs per member using Adam, learning rate 10^{-3} and weight decay 10^{-4}. The downstream networks use the same optimizer settings for 200 epochs. Upstream checkpoints minimize the grouped DEV objective above; downstream and direct checkpoints minimize DEV Brier score. These different selection objectives are declared rather than treated as interchangeable. All three members use the same data split, so they are optimization realizations, not independent experimental datasets.

V. Candidate Evaluation and Execution

For jump candidates, complete-task probability is estimated by

$$\widehat{Q}(s,a) = \widehat{P}(V_h \mid s,a)\int R_\phi(z)\widehat{p}_\theta(z \mid V_h, s, a)\, dz. \quad (8)$$

The implementation averages this estimate over the three upstream members and 32 samples per member. Common Gaussian draws are used across candidates within a query. Sampled quaternions are normalized, sampled handoff time is bounded below by query time, and the latest-frame validity flag is set to one. These operations enforce representation conventions; they do not ensure dynamically realizable samples. Ensemble spread is not used as a certified risk margin.

Ground-only candidates are scored with p_g. No additional contact probability is multiplied into Eq. (8): the full-goal target already requires the relevant preceding physical events. The decomposition defines an estimator, not automatic probability calibration.

The planner evaluates 60 candidates: four route types, five speeds, and three preferences. The routes are straight crossing, S-bend crossing, right-angle crossing, and ground detour. Available speeds are 1.5, 1.8, 2.0, 2.3, and 2.6 m/s, with $\alpha \in \{0, 0.5, 1\}$. Route geometry and geometric admission are shared across methods. The highest-scoring admitted candidate is selected. Actual computation time is rounded up to a control interval while existing control continues, followed by a publication-time admission recheck. No physical state is restored to the query state.

The present implementation performs one moving query and a publication recheck. It does not repeatedly replan from later measured states. Postlanding references remain fixed, and neither controller receives a new learned policy during these experiments.

VI. Experimental Design

A. Data and grouped fitting

The initial collection contains 720 executions, partitioned by eight TRAIN and four DEV starting conditions. A second collection requests 1,440 executions at multiple query positions, of which 1,199 are published and attributable to the selected candidates. The 241 unpublished requests remain in the original record but are not assigned counterfactual candidate outcomes. This second collection uses six TRAIN and two DEV roots. Across the combined 20 roots, no exact initial-state duplicate crosses the split.

The resulting upstream dataset contains 1,379 TRAIN executions with 205 goals and 540 DEV executions with 79 goals. Valid handoffs provide 463 TRAIN and 168 DEV examples for the downstream model. All observed candidate failures remain in the corresponding classification denominators. A further 240 executions from four previously inspected roots provide historical transfer analysis, not an untouched confirmation set.

An initial model trained only at the early query position had a historical multi-position handoff-position RMSE of 2.32 m. Including the previously partitioned distance-training data was a declared development change. Neither that earlier fit nor its failure is discarded.

B. Online panel and comparators

Before online execution, model weights and 30 newly sampled starting conditions were frozen. Ten conditions are assigned to each query position $x \in \{1.8, 3.0, 5.0\}$ m. Initial lateral offset, yaw, and roll vary within ± 0.03 m, ± 0.035 rad, and ± 0.02 rad, respectively; initial speed lies in $[1.9, 2.1]$ m/s. Each condition is evaluated three times per method, crossing methods over environment slots. Thus each method has 90 episodes but only 30 independent starting conditions.

The direct-success comparator uses the same 582-dimensional query input, grouped data, hidden widths, and three-member fitting budget, with a scalar goal output. The distance rule selects fixed candidates derived from the earlier TRAIN data for each query position. A TRAIN-only route-type completion prior and a constant completion prior are additional offline baselines. Controllers, physical task limits, and the ten-second observation horizon are common.

The primary outcome is complete-task success. Road violations and computation time are reported separately. Paired gains and losses include all matched condition–slot pairs. Confidence intervals resample starting conditions, stratified by query position, with 5,000 bootstrap samples; the three repetitions are averaged within each condition. Recorded query root states are compared across methods. Equality of these records does not certify equality of hidden PhysX solver state.

TABLE I

Completion Brier scores. DEV participates in checkpoint selection; historical transfer is previously inspected.

Predictor	DEV (540)	Historical (240)
Two-interface	.11520	.11139
Direct success	.12336	.11839
Route-type prior	.11673	.10998
Constant prior	.12490	.12165

TABLE II

Conditional goal Brier by input subset. Historical sets were previously inspected; DEV selects epochs.

Inputs	DEV 168	Source 65	RB 70
Time only	.21142	.18155	.18660
Geometry/speed/time	.19686	.14725	.15194
With dynamics	.14450	.13305	.13488
Root/time	.14178	.12881	.13602
Root/frame/time	.14502	.12531	.13610

VII. Results

A. Physical interface and prediction

The earlier stage audit found both approach and continuation failures. In the 720-execution collection, 265 executions did not reach jump entry; among 217 handoffs, 153 subsequently failed. Of those handoffs, 26 had already violated the road constraint. In the 1,199 published distance executions, 383 did not reach entry and 379 of 493 handoffs subsequently failed, with 53 prior road violations. These counts motivate both interfaces; they do not support assigning the entire bottleneck to landing recovery.

The interface estimator has lower DEV Brier score than the direct comparator, but the route prior is competitive and has the lowest historical-transfer score (Table I). Conditional handoff-position RMSE is 0.1317 m on DEV and 0.1299 m on historical transfer, computed over Cartesian components of valid handoffs. Handoff-time MAE is 0.3711 and 0.4588 s, respectively. These conditional errors exclude absent handoffs; they cannot replace occurrence evaluation or full-task outcomes.

B. Information in the handoff state

To test whether the continuation model uses more than elapsed time and nominal progress, we fitted three feature subsets on the same 463 TRAIN and 168 DEV valid handoffs. Each uses three 64–32 SiLU networks, 200 epochs, and the same grouped DEV checkpoint criterion. The geometric subset contains world x, y, mount-corrected yaw, forward velocity, and time. The dynamics extension adds roll, pitch, lateral and vertical velocity, and three world angular velocities. Existing root-only and latest-frame models are reused unchanged. Adding dynamics lowers Brier relative to geometry/speed/time in all six DEV roots and both historical sets (Table II). Time alone performs near the constant prior.

TABLE III

Actual complete tasks. Each cell at a query position contains ten independent roots with three repetitions.

Query x (m)	Interface	Direct	Distance
1.8	0/30	11/30	30/30
3.0	10/30	13/30	9/30
5.0	5/30	9/30	2/30
Total	15/90	33/90	41/90
Road violations	9/90	13/90	22/90
Mean query (ms)	6.402	3.506	2.653

Thus measured handoff dynamics contains predictive information in the fixed downstream context. This is a grouped feature diagnostic, not an isolated causal effect of lean: several state fields and input parameter counts change together. The latest observation frame does not consistently improve on the physical root alone. Predictive information at observed handoffs does not establish that sampling those states upstream improves candidate selection.

C. Executed decisions

Table III shows a clear overall deficit. The interface model gains 13 and loses 39 matched episodes relative to the rule. Its mean condition-level completion difference is -0.289, with a root-bootstrap 95% interval of $[-0.400, -0.178]$. Against direct success, it gains five and loses 23; the difference is -0.200, with interval $[-0.344, -0.067]$. All 90 paired query root states are exactly equal in each comparison. The deficit therefore cannot be dismissed as a mismatch in the recorded query states.

The interface completes 5/30, 3/30, and 7/30 across the three slot rotations; the rule completes 13/30, 14/30, and 14/30. The full panel uses 1,080,000 physical environment steps and 268.06 s summed simulation wall time. Mean query times are below one control period, but a 20-ms physical publication delay is still propagated. This measured result does not establish an architecture-independent latency bound.

D. Why prediction quality did not yield useful ranking

The interface selects 33 detours, all at 1.5 m/s and $\alpha = 0$, and all time out. The other 57 selections are straight crossings. In TRAIN alone, this detour candidate has zero goals in 20 executions yet a mean predicted completion probability of 0.238. A 1.8 m/s, $\alpha = 0$ detour has 16/20 goals and mean predicted probability 0.287. The ground head therefore poorly separates useful actions even before online evaluation.

The shared representation and joint checkpoint objective may contribute to this failure. A bounded follow-up independently fitted a ground-only network with the same hidden widths and 250-epoch budget, selecting by ground DEV Brier. Ground DEV Brier improved from 0.16439 to 0.15916, but historical ground-transfer Brier

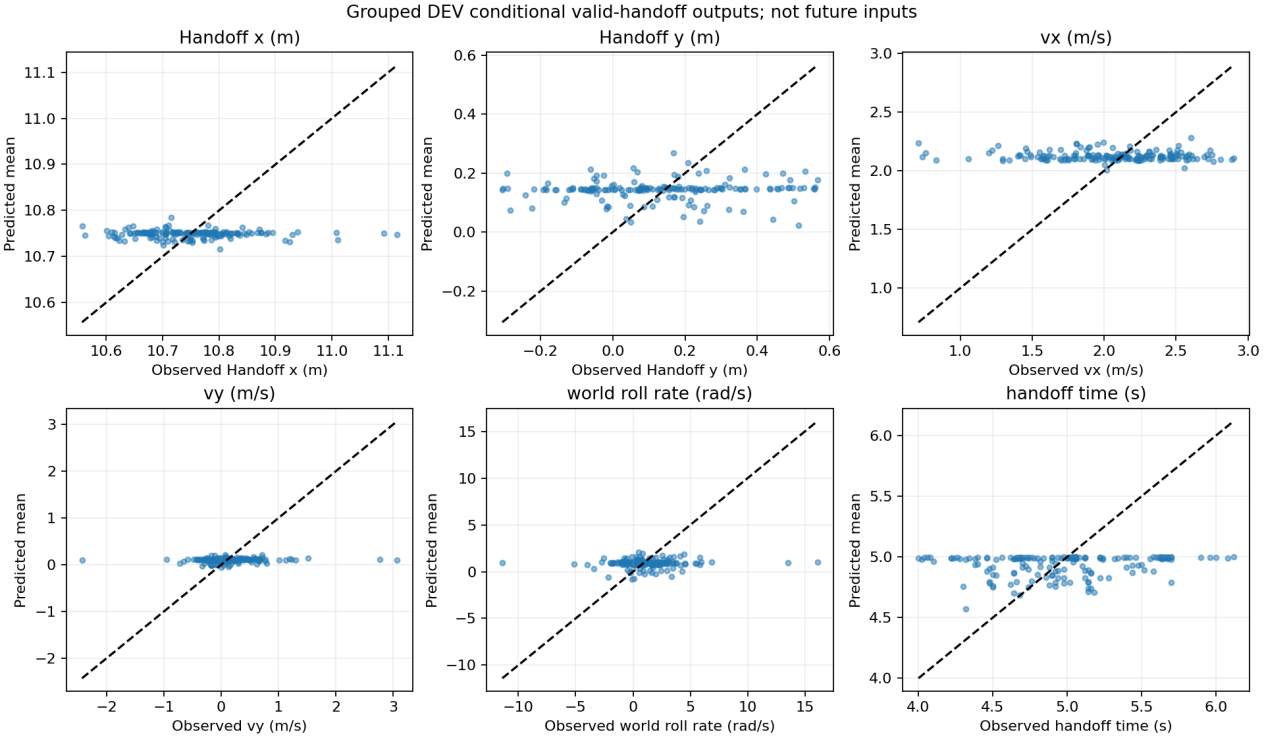


Fig. 1. Predicted and observed handoff outputs on 168 valid DEV handoffs. State regression is conditional on valid handoff; the event classifier also includes failed and missing handoffs. The plots expose physical prediction errors rather than only a final completion score.

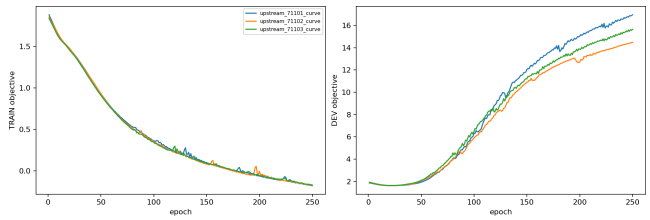


Fig. 2. Recorded upstream/direct training and development curves. Upstream selection uses the masked joint objective; the direct comparator uses DEV Brier. Their numerical scales are not interchangeable.

worsened from 0.20611 to 0.20728. The zero-success low-speed candidate remained overestimated (TRAIN mean 0.34432). This correction therefore does not establish that the shared objective is the sole cause. These offline follow-up results are separate from the frozen online panel in Table III. A subsequent development retest on the same 30 roots, with the same three slot rotations, completes 14/90 with the independent ground head, 30/90 with direct success, and 42/90 with the distance rule. All 29 selected detours still time out. Only four interface decisions change relative to the original panel. The unchanged comparators also vary between batches, so the 15-to-14 difference alone is not a causal estimate of fitting effects. The dominant ground-route failure remains unresolved. Aggregate Brier scores average over many unsuccessful candidates and do not directly evaluate the top-ranked choice. The modest gains at middle and late positions cannot compensate for the early-query loss. Fewer road violations likewise do not imply greater task success when the selected motion times out.

VIII. Discussion

The two-interface formulation exposes what the planner predicts: whether a lawful handoff will occur, the state and time delivered at that handoff, and the conditional outcome under a specified downstream controller. It removes the need to treat a nearly deterministic controller-switch event as a learned intermediate hazard. It also avoids multiplying marginal contact and goal probabilities whose failure events overlap. These structural choices improve the definition of the estimator, but the online results show that they do not establish a planning advantage.

Several boundaries remain. The conditional model is trained under fixed cold initialization and a fixed downstream straight road. It does not measure a recoverability margin or distinguish states that can be rescued by a short correction. A nominal ground-motion envelope was examined separately, but many successful two-second continuations did not enter it; envelope membership is therefore not adopted as a necessary capture condition. Measured initialization also failed to improve an eight-source same-slot diagnostic, which does not rule out useful initialization from genuine controller histories.

The relationship to skill repair remains a hypothesis requiring matched interface evidence. Earlier whole-task substitution yielded 34/240 completions with the original jump skill and 35/240 with RB400. This denominator correctly measures system-level utility, including failures before the skill is activated. Conditional skill comparisons

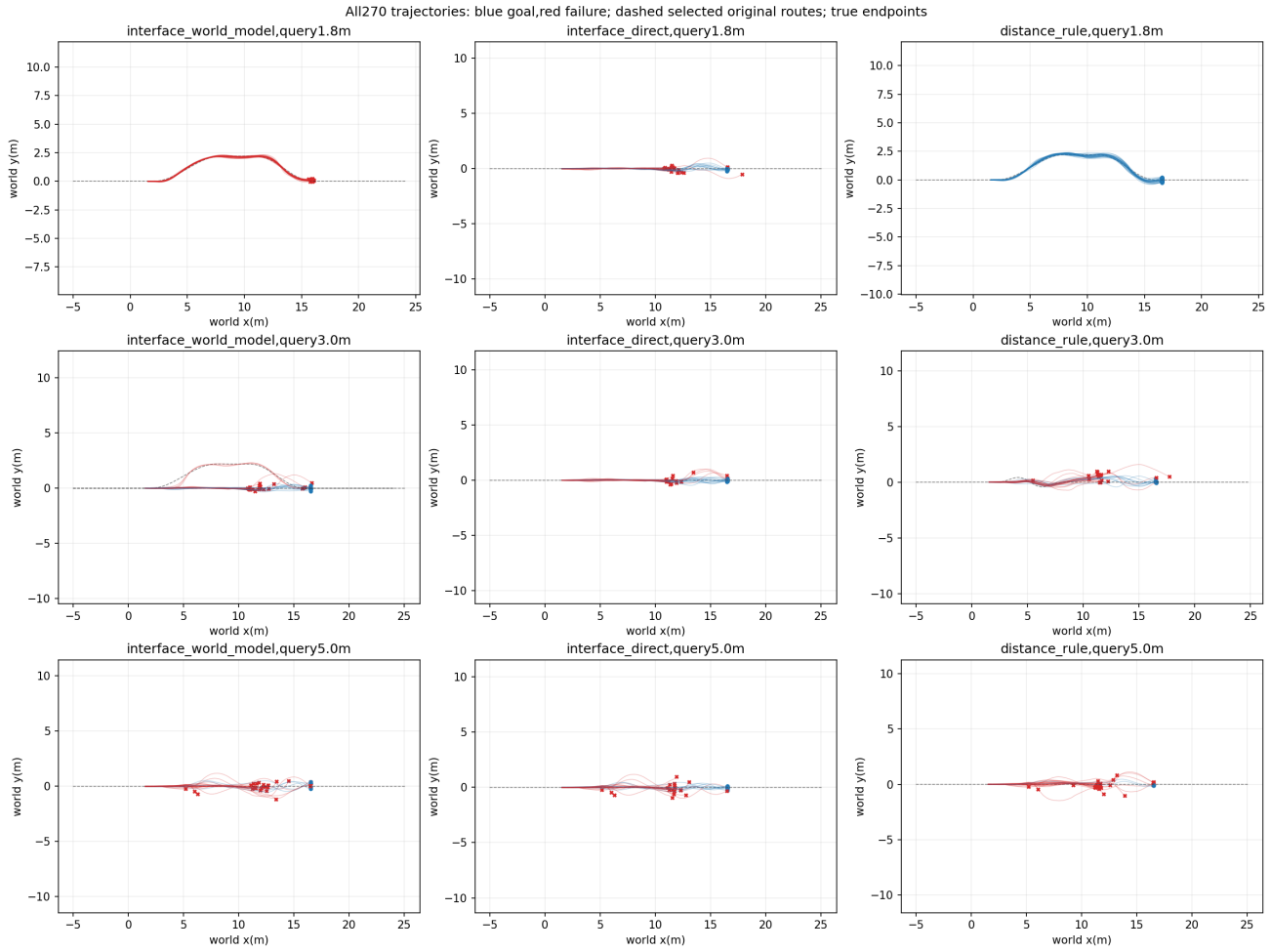


Fig. 3. All 270 actual XY executions and selected reference routes, grouped by query position and method. Successful and failed trajectories are retained and terminate at their recorded endpoints. Each method has 30 episodes per position from ten independent conditions with three slot-crossed repetitions.

at matched legal entrances answer a different question and should supplement, rather than replace, the whole-task result.

The immediate methodological gap is candidate ranking, especially the ground branch. The independent ground-head fit and development retest leave the dominant failure unresolved without changing task deadlines or retraining the jump skill. If the present 30 roots inform that correction, they become development evidence for the repaired model. Broader tests should vary downstream corridor geometry and evaluate new measured-state decisions. Neither capability is claimed here, and no hardware or new-map generalization is established.

IX. Conclusion

We implemented a handoff-conditioned world model for composing frozen driving and jumping controllers on a known off-road map. The model predicts a physical handoff distribution and evaluates its downstream consequences, with explicit training targets and a finite-candidate execution interface. The new online panel confirms completed jump–ground tasks but shows lower completion than both a direct classifier and a distance rule. A poorly discriminating ground-route head accounts for a substantial observed failure mode. The study provides a reproducible interface and a concrete diagnosis; reliable task-conditioned recovery prediction and improved candidate selection remain open.

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